Light

GCSE Computer Science — Programming Data Structures (Set 2)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What happens when this line of code executes?

```
age = 16
```

- **A.** It checks if age equals 16
- **B.** It prints the number 16 on screen
- **C.** It creates a constant that cannot be changed
- **D.** It creates or updates a variable called age and stores the value 16 in it

Q2 (1 mark)

Which of the following is the best example of naming a constant?

- A. maxSpeed
- B. MAX_SPEED
- C. max speed
- D. Max-Speed

Q3 (1 mark)

Which data type should be used to store a person's full name?

- A. Integer
- B. Boolean
- C. Character
- D. String

Q4 (1 mark)

A weather application needs to store temperature readings like 18.5°C. Which data type is most appropriate?

- A. Real
- B. Integer
- C. String
- D. Boolean

Q5 (1 mark)

A program needs to track whether a user has agreed to the terms and conditions. Which data type should be used?

- ○ **A.** String
- ○ **B.** Integer
- ○ **C.** Boolean
- ○ **D.** Real

Q6 (1 mark)

A teacher needs to store the test scores of 30 students. Why is an array better than using 30 separate variables?

- ○ **A.** Arrays use more memory, making the program faster
- ○ **B.** Arrays can only store numbers, which is what test scores are
- ○ **C.** Arrays automatically calculate the average of all values
- ○ **D.** Arrays allow all related data to be stored under one name and processed using loops

Q7 (1 mark)

An array is declared as: `colours = ['Red', 'Green', 'Blue', 'Yellow']`
What value is stored at index 2?

- ○ **A.** Red
- ○ **B.** Green

- ○ **C.** Blue
- ○ **D.** Yellow

Q8 (1 mark)

Given `name = 'JAMES'`, what is the output of `name.lower`?

- ○ **A.** JAMES
- ○ **B.** James
- ○ **C.** 5
- ○ **D.** james

Q9 (1 mark)

Why might a programmer use `str(42)`?

- ○ **A.** To convert the number 42 into the text '42' for concatenation with other strings
- ○ **B.** To round the number 42 to the nearest ten
- ○ **C.** To check if 42 is a valid number
- ○ **D.** To store 42 in an array

Q10 (1 mark)

Given `password = 'abc123'`, what does `password.length` return?

- ○ **A.** 3
- ○ **B.** 6
- ○ **C.** 123
- ○ **D.** abc

Q11 (1 mark)

A program needs to store a student's grade as a single letter (A, B, C, D, or F). Which data type is most memory-efficient?

- ○ **A.** Character
- ○ **B.** String
- ○ **C.** Integer
- ○ **D.** Boolean

Q12 (1 mark)

Why would a programmer use a record data structure?

- ○ **A.** To store a single value that changes frequently
- ○ **B.** To repeat a block of code multiple times
- ○ **C.** To group related data of different types together under one name
- ○ **D.** To sort data into alphabetical order

Q13 (2 marks)

An array is declared as: `days = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri']`
What happens if the program tries to access `days[5]`?

- ○ **A.** It returns 'Fri'
- ○ **B.** It returns an empty string
- ○ **C.** It returns 'Mon' by wrapping around to the start
- ○ **D.** It causes an index out of bounds error because valid indices are 0-4

Q14 (2 marks)

Given `postcode = 'SW1A 2AA'`, what is the output of `postcode.left(4)`?

- ○ **A.** SW1A
- ○ **B.** SW1A 2AA
- ○ **C.** 2AA
- ○ **D.** 2AA

Q15 (2 marks)

What is the output of the following code?

```
price = '50'
quantity = 3
total = int(price) * quantity
print(total)
```

- ○ **A.** '505050'

- B. 150
- C. '150'
- D. Error: cannot multiply string by integer

Q16 (2 marks)

A 2D array is declared as:

```
seating = [['A1', 'A2', 'A3'],
           ['B1', 'B2', 'B3']]
```

How many rows and columns does this array have?

- A. 3 rows, 2 columns
- B. 6 rows, 1 column
- C. 1 row, 6 columns
- D. 2 rows, 3 columns

Q17 (2 marks)

Given `filename = 'report.pdf'`, which operation extracts 'pdf'?

- A. `filename.left(3)`
- B. `filename.substring(0, 3)`
- C. `filename.right(3)`
- D. `filename.upper`

Q18 (2 marks)

Given `scores = [10, 20, 30, 40]`, what does the array contain after executing `scores[1] = 25`?

- A. [25, 20, 30, 40]
- B. [10, 25, 30, 40]
- C. [10, 20, 25, 40]
- D. [10, 20, 30, 25]

Q19 (2 marks)

What is the result of `float('7') / 2`?

- A. 3
- B. 3.5
- C. '3.5'
- D. Error

Q20 (2 marks)

A program has variables: `firstName = 'Emma'` and `lastName = 'Watson'`
Which expression produces 'Emma Watson'?

- A. `firstName + ' ' + lastName`
- B. `firstName + lastName`

- C. `firstName - lastName`
- D. `firstName.lastName`

Q21 (2 marks)

A 2D array stores a noughts and crosses grid:

```
grid = [['X', 'O', 'X'],
        ['O', 'X', 'O'],
        ['X', 'O', 'X']]
```

What value is returned by `grid[2][0]`?

- A. O
- B. Error - invalid index
- C. The entire third row
- D. X

Q22 (2 marks)

A record is defined as:

```
student record:
  name: string
  age: integer
  grade: character
```

A variable `pupil` is created using this record. How would you access the student's age?

- A. `pupil[age]`
- B. `pupil.age`
- C. `age.pupil`

- D. `record.age`

Q23 (2 marks)

A variable is declared inside a function. Where can this variable be accessed?

- A. Only within that function (local scope)
- B. Anywhere in the entire program (global scope)
- C. Only in functions called after this function
- D. Only in the main program, not in functions

Q24 (3 marks)

A program needs to find the position of a specific value in an unsorted array of 100 elements.
Analyse why a linear search must be used rather than a binary search.

- A. Binary search requires the array to be sorted; since this array is unsorted, linear search is the only option - it checks each element sequentially until the value is found or all elements are checked
- B. Binary search is faster, so it should be used instead
- C. Linear search requires the array to be sorted first
- D. Both searches work equally well on unsorted data

Q25 (3 marks)

An email validation algorithm needs to check that an email address contains exactly one '@' symbol.

Analyse how string operations could implement this check.

- ○ **A.** Use .length to count how many '@' symbols are present
- ○ **B.** Use .upper to convert the email then check for '@'
- ○ **C.** Iterate through each character using a loop with .substring(i, 1), incrementing a counter each time '@' is found; after checking all characters, verify the counter equals exactly 1
- ○ **D.** Use .left(1) to check if the first character is '@'

Q26 (3 marks)

A cinema booking system stores seat availability in a 2D array where True means available and False means booked:
`seats[5][10]` represents 5 rows of 10 seats each.
Analyse how to count the total number of available seats.

- ○ **A.** Use seats.length to get the count of True values
- ○ **B.** Use nested loops: outer loop iterates through rows (0 to 4), inner loop iterates through columns (0 to 9), incrementing a counter whenever seats[row][col] equals True
- ○ **C.** Check only seats[0][0] and multiply by the total number of seats
- ○ **D.** Add all the values in the array together

Q27 (3 marks)

A music streaming service needs to store information about songs: title, artist, duration (in seconds), and whether it's explicit content.
Analyse the most appropriate data structure design.

- ○ **A.** Four separate variables for each song

○ **B.** A single string containing all song information separated by commas

○ **C.** A record structure with fields: title (string), artist (string), duration (integer), explicit (Boolean); multiple songs stored in an array of these records

○ **D.** A 2D array with rows for each song and columns for each piece of information

Q28 (3 marks)

A program receives user input for their age. The input arrives as a string. Analyse the potential problems with using `int(userInput)` directly.

○ **A.** There are no potential problems; int() always works correctly

○ **B.** int() will round decimal numbers incorrectly

○ **C.** int() only works with single-digit numbers

○ **D.** If the user enters non-numeric text (like 'twenty' or 'abc'), the cast will fail and crash the program; input validation should check that the string contains only digits before attempting to cast

Q29 (3 marks)

A program needs to find the largest value in an array of exam scores. Analyse the correct algorithm structure.

○ **A.** Compare the first and last elements only

○ **B.** Sort the array and return the first element

○ **C.** Add all elements together and divide by the array length

○ **D.** Initialise a variable 'highest' with the first element, then iterate through the remaining elements; whenever an element is greater than

'highest', update 'highest' to that value; after checking all elements, 'highest' contains the maximum

Q30 (3 marks)

A program receives a full name in the format 'FirstName LastName' (e.g., 'John Smith') and needs to extract just the last name.
Analyse how string operations could achieve this.

- ○ **A.** Use .right(5) to get the last 5 characters
- ○ **B.** Find the position of the space character, then use .substring() to extract everything after the space; alternatively, calculate the space position and use .right() with the remaining length
- ○ **C.** Use .upper to convert to uppercase then extract
- ○ **D.** Use .length to count the characters in the last name

Total: **55** marks • Time: **30 min**

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